

Section 1: Project Information

Project Name: S.W.A.G. (Smart Wear AI Guide)

Semester: Fall 2025

Group Number: 6

Project Coordinator: Ashir Imran

Guide: Dr. Tushara Sadasivuni

Team Members:

- Harshini Mahesh
 - Ashir Imran
 - Udit Chandrashekar
 - Jinseon Ryu
 - Irfan Bhuiyan
 - Khoi Dang
 - Date of Submission: 9/11/2025
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Section 2: Teamwork Basics

Ground Rules (Norms 1–5)

1. Respect everyone's ideas. Example: In my CS4520 group, everyone shared project ideas before we voted, which encouraged quieter teammates.
2. Be on time and prepared. Example: In my Networking project, we agreed to join meetings with updates ready, which saved time.
3. Divide tasks fairly. Example: In Hacklytics, we split backend/frontend tasks, which avoided overlap.
4. Communicate clearly. Example: For Software Engineering diagrams, we used Google Docs for updates, so no one was confused.
5. Be accountable. Example: I once missed a UML deadline but informed my group so they could adjust—this to keep trust intact.

Hints for Handling Difficult Behavior

- If someone dominates, rotate who speaks first. In my Data Science group, each person had to present one slide.
- If someone is disengaged, assign them a key but smaller task. In Algorithms, we gave a quiet member the testing role, and they became engaged.

Hints for Handling Group Problems

- Missed deadlines: Build buffer days. In Networking, we submitted drafts 2 days early as backup.
- Disagreements: Use majority vote. On the PSA board, we voted on food vendors to avoid long debates.
- Lack of quality: Pair with stronger ones. In Flutter, I paired with a teammate new to Dart so we could both finish successfully.

Section 3: Brief Resumes

(Each member adds 3–4 lines. Example below for Ashir others fill in their own.)

Ashir Imran – Computer Science major at Georgia State University. Experience with FastAPI, Flutter, and GitHub. Projects include Hacklytics 2025 (AI health monitoring) and Atlantex Apparel e-commerce backend.

Harshini's resume – Languages: Python, Java, C++, Bokeh, SeaBorn, SQL, HTML. Data Analytics and Automation Intern & CS Student Research Assistant.

Udit's resume - Languages: Python, Java, C++, RISC-V Assembly, ARM64, HTML, SQL. Tools & Frameworks: Git/GitHub, Visual Studio Code, Unix/Linux CLI, Figma, Postman. Soft Skills: Leadership, Problem Solving, Communication, Team Collaboration

Jinseon's resume - Computer Science major at Georgia State University. Experience with Python, C++, Java, and HTML. Projects include Eldaid (Senior supporting device) and performance-based level progression database.

Irfan's resume - Computer science major at Georgia State. Experience with flutter, dart, and git. Well versed in python, java, css, and HTML. Projects include an employee management system and smart expense tracker.

Khoi's resume- Computer science major at Georgia State. Languages: Python, Java, C++, C#. Have experience in computer vision, including OpenCV. CompTIA Network++ certified.

Section 4: Scheduling and Planning Table for A1

Task	Responsible	Start Date	End Date	Status
Teamwork Basics Summary	Harshini	09/10	09/12	Pending
Brief Resumes	All members	09/10	09/12	Pending
Problem Statement Draft	Ashir	09/11	09/14	Pending
Context Diagram	Udit	09/11	09/14	Pending
Activity Diagram	Jinseon	09/12	09/15	Pending
Final Report Assembly	Irfan	09/15	09/16	Pending

Section 5: Problem Statement

Product (high level):

S.W.A.G. (Smart Wear AI Guide) is a webapp that acts as a virtual closet and AI stylist, helping users decide what to wear based on their clothes, the weather, and location.

Whom is it for?

Students, professionals, and style-conscious individuals who want quick, weather-ready outfit suggestions.

What problem does it solve?

Saves time choosing outfits, prevents repeating looks too often, and ensures clothes match weather conditions.

Alternatives:

Apps like Smart Closet and Pureple exist, but they lack weather integration, AI-powered recs, and calendar planning.

Why compelling?

It combines daily weather checks (something users already do) with personal closet management, making it both practical and engaging.

Objectives & differentiators:

- Weather-aware recommendations.
- AI stylist chatbot.
- Calendar outfit planning with history.
- Differentiator: Weather + closet + AI chatbot + 3D demo.

Competitors vs novelty:

Competitors manage closets, but they don't integrate real-time conditions or AI learning.

Our novelty: forecast-driven outfit choices and personalized styling.

Feasibility:

Built with Next.js, PostgreSQL, weather APIs, ML APIs for tagging clothes. Cloud hosting (Vercel/Supabase) ensures it runs easily.

Interesting technical points:

- Real-time weather integration.
- AI embeddings for clothing similarity.
- Chat-based recommendation engine.

Logins:

Yes — Client login (user's closet + AI recs) and Admin login (manage taxonomy, monitor system).

Section 6: System Requirements

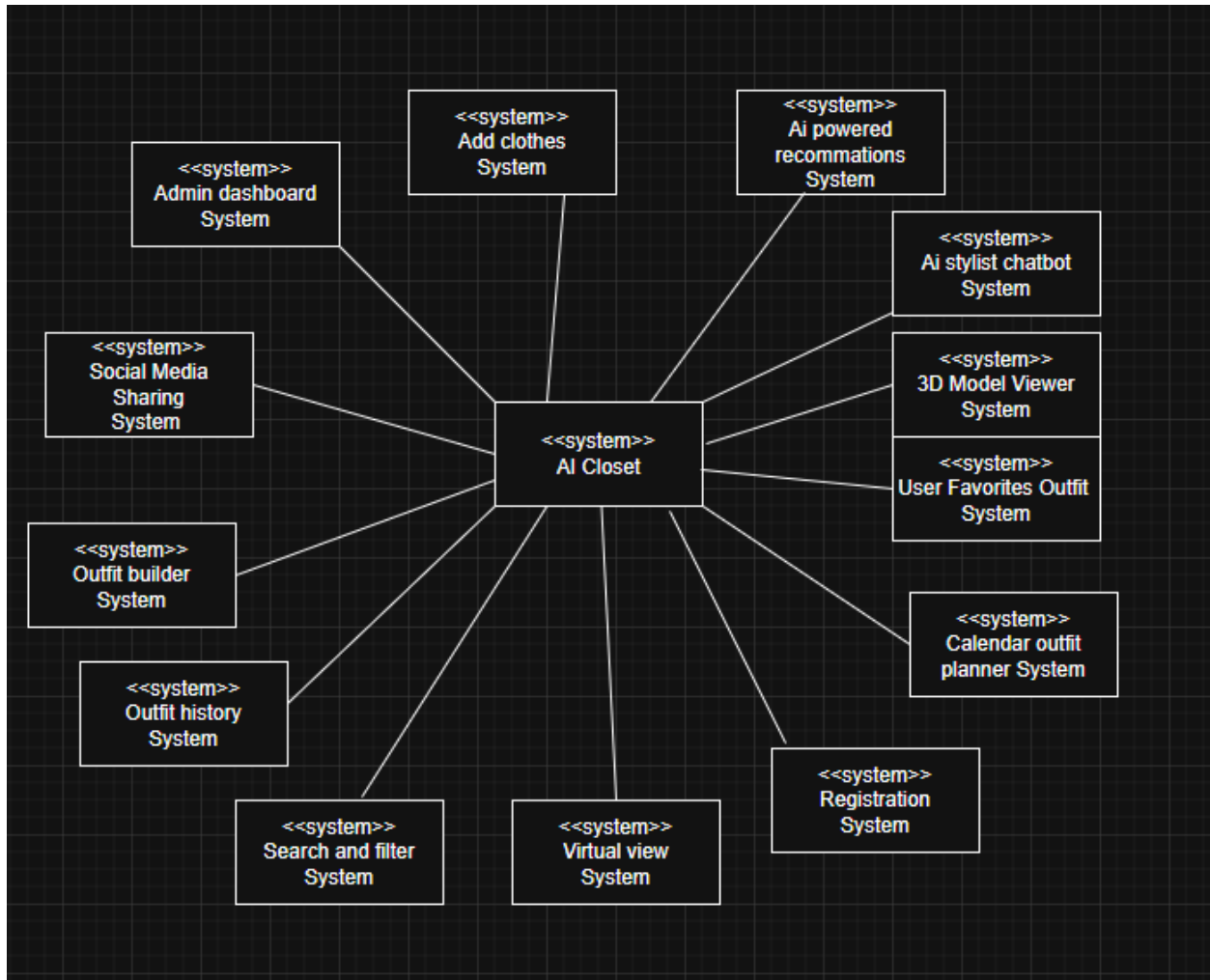
Context Diagram (high-level description)

- Actors: User, Weather API, AI Image Service, Database, Admin.
- Flow:
 - User uploads clothing → system tags → stored in DB.
 - User requests outfit → system checks closet + weather API + rec engine → returns outfit.

- Admin monitors taxonomy & system logs.

(Diagram: system in center; arrows from User, Weather API, DB, Admin → system → back to User.)

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Activity Diagram (flow description)

1. User logs in.
2. Uploads clothing → system runs background removal & tagging → saves in DB.
3. User requests “Today’s Outfit.”
4. System fetches closet + weather forecast + past outfits.
5. AI generates 3 outfit suggestions.
6. User selects outfit → marks as worn.

7. History updates: preferences refined for future recs.

(Diagram: swim lanes → User & System; arrows showing above steps.)

